땀 속 포도당 및 케톤 모니터링을 위한 V_2CTx-MXene@f-MWCNTs 표피 패치

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A V_2CTx-MXene@*f*-MWCNTs Epidermal Patch for Sweat Glucose and Ketone Monitoring

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Abstract

Continuous, non-invasive monitoring of metabolic biomarkers such as glucose and β-hydroxybutyrate (HB) in human sweat is crucial for diabetes management, ketoacidosis risk detection, and ketogenic diet monitoring. While continuous glucose monitoring has advanced significantly, HB sensing in sweat remains largely unexplored. Here, we report a microfluidic channel-integrated multifunctional wearable electrochemical biosensing patch capable of simultaneously monitoring glucose and HB continuously. A synergistic V_2CTx@*f*-MWCNTs composite enhances electroactive surface area, suppresses MXene restacking, and improves electron transfer. A diffusion-limiting layer further improves HB linearity, stability, and interference resistance. The patch integrates glucose, HB, pH, and temperature sensors with real-time physiological compensation. The device achieved high sensitivities of 131.41 μA mM^{-1} cm^{-2} for glucose and 53.81 μA mM^{-1} cm^{-2} for HB, demonstrating stable operation for ~171 minutes.

Keywords: ***MXene@MWCNT Nanocomposite, Glucose, β-Hydroxybutyrate, multifunctional Patch***

1. Introduction

Non-invasive monitoring of metabolic biomarkers has accelerated the development of wearable biosensors, with sweat emerging as a valuable biofluid for real-time health assessment [1]. Among sweat biomarkers, glucose and β-hydroxybutyrate (HB) are critical for managing diabetes, ketoacidosis, and ketogenic diets, yet their low concentrations challenge reliable detection [2]. Although electrochemical biosensors provide high sensitivity, their performance depends strongly on the transduction layer. Conventional carbon materials suffer from aggregation, hydrophobicity, and limited stability, restricting long-term operation. HB sensing is further complicated by interference, diffusion limitations, and nonlinear electron transfer under low sweat flow [3]. Here, we present a microfluidic-integrated wearable biosensor based on a V_2CTx@*f*-MWCNTs–laser-induced graphene hybrid. The architecture enhances electron transfer, suppresses aggregation, and enables stable, multiplexed monitoring of glucose, HB, pH, and temperature.

2. Fabrication method

This study utilized analytical-grade chemicals, enzymes, and polymers to fabricate a multifunctional wearable electrochemical biosensor. V_2AlC MAX phase, functionalized multi-walled carbon nanotubes (*f*-MWCNTs), glucose oxidase, β-hydroxybutyrate dehydrogenase, and redox mediators TBO served as key sensing components. Flexible laser-induced graphene (LIG) electrodes were patterned on polyimide substrates using a CO_2 laser. V_2CTx MXene was synthesized via mild HF/HCl etching, delaminated, and integrated with *f*-MWCNTs to form a conductive nanohybrid deposited on LIG [4]. Glucose and HB sensors were fabricated using enzyme immobilization, mediators, and diffusion-limiting layers. Additional pH and temperature sensors and PET-based microfluidic channels enable real-time sweat analysis.

3. Result and discussion

The structural characterization and sensing performance of a microfluidic-integrated multifunctional sweat biosensing patch based on V_2CTx@*f*-MWCNTs modified laser-induced graphene (LIG). FESEM confirmed successful synthesis multilayer V_2CTx MXene and revealed uniform anchoring of COOH-functionalized MWCNTs within MXene interlayers. An optimized V_2CTx@*f*-MWCNT (1:1) ratio produced a homogeneous conductive network [5]. Raman, XRD, and XPS verified MXene formation, surface terminations (–O/–OH/–F), partial residual MAX phase, and enhanced carbonaceous interactions, indicating improved conductivity. Electrochemical analysis showed the optimized hybrid exhibited the highest redox currents, lowest charge-transfer resistance (~99 Ω), and largest electrochemical surface area. The glucose sensor achieved high sensitivity (~133 μA mM^{-1} cm^{-2}, $R^2 \approx 0.999$) with excellent stability and selectivity. The HB biosensor delivered a linear response (0.01–0.5 mM), sensitivity ~53.8 μA mM^{-1} cm^{-2}, and LOD of 10 μM. Integrated microfluidics enabled real-time multiplexed monitoring of sweat glucose, HB, pH, and temperature with on-body validation.

4. Conclusion

We developed A microfluidic-integrated wearable electrochemical biosensor using a V_2CT_x@f-MWCNT/LIG nanocomposite was developed for real-time, noninvasive monitoring of glucose and β-hydroxybutyrate (HB) in sweat. The synergistic nanocomposite enhanced conductivity, electroactive surface area, and stability, yielding high sensitivity and reproducibility. Integrated diffusion control, pH and temperature calibration, and microfluidics ensured reliable sensing under dynamic conditions, demonstrating strong potential for scalable, personalized wearable health-monitoring systems.

Acknowledgments

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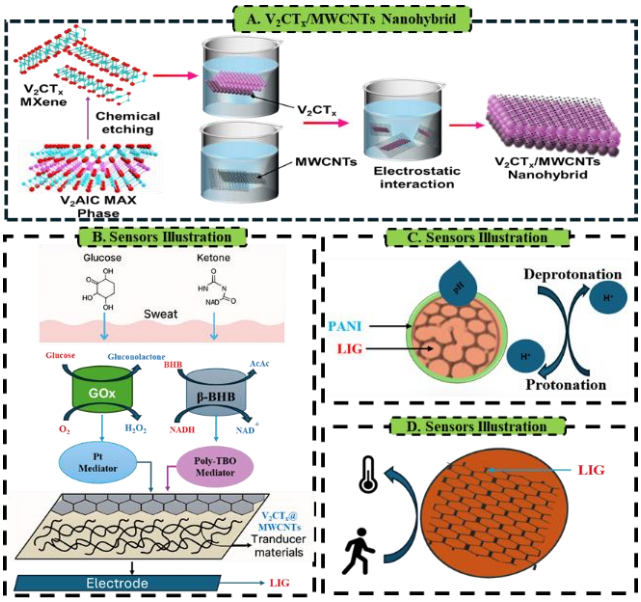

Fig. 1. A) Schematic illustration of the Nanohybridization of V_2CTx@*f*-MWCNTs B) Functionalization of the Glucose, and ketone sensor C) pH biosensor sensor fabrication. D) Temperature sensor fabrication.

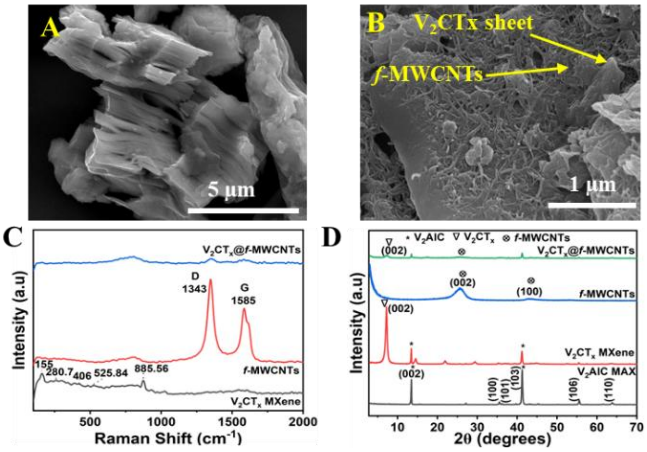

Fig. 2. A) FESEM images of V_2CTx B) V_2CTx@MWCNTs nanocomposite, C) Raman spectra D) XRD patterns of nanocomposite.

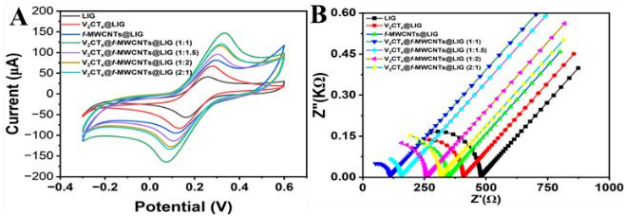

Fig. 3. A) and B) Cyclic Voltammetry (CV) and EIS fitted curves for the optimization of electrodes.

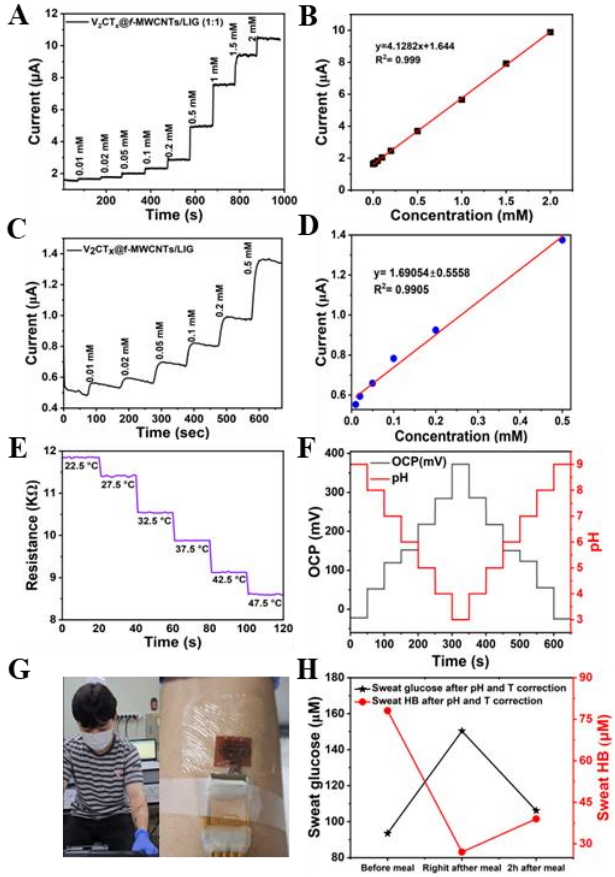

Fig. 4. A) & B) Amperometric response and Linearity fitting curve of the glucose biosensor. C) & D) Amperometric response and Linearity fitting curve of the HB sensor. E) In the physiological temperature range, resistive responses of the LIG based temperature sensor. F) The OCP versus time curves of the pH sensor, as well as repeatability test. G) The multi-modal patch worn on the subject's forearm during exercise on a cycle ergometer. H) Relation between glucose and HB in sweat at different conditions.

Table 1: Performance of the multi-functional patch

Biomarker	Sensitivity ($\mu AmM^{-1}cm^{-2}$)	Linearity	Linear Range	LOD
Glucose	131.41	0.999	0.002-2.00 (mM)	1(μM)
HB	53.81	0.991	0.01–0.5 (mM)	10 (μM)
pH	62.32	0.992	pH: 3-9	-
Temperature	-0.062 (%˚C^{-1})	0.997	22.5-47.5 (˚C)	-